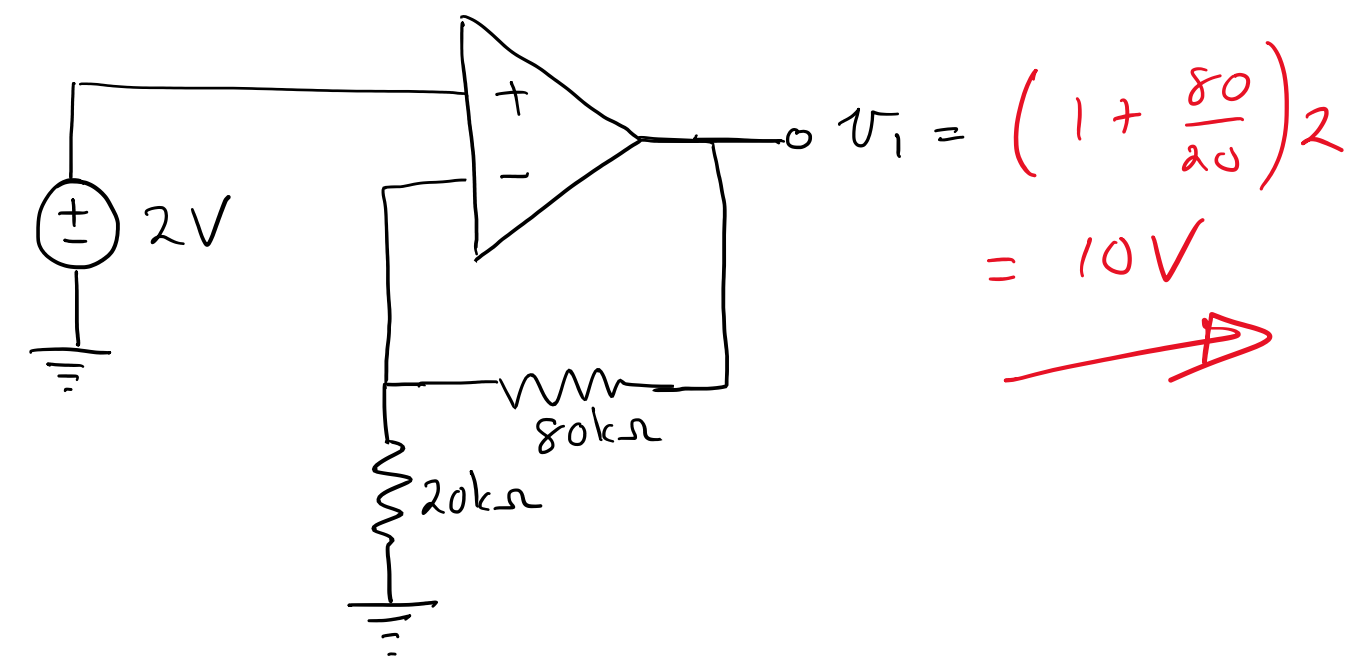
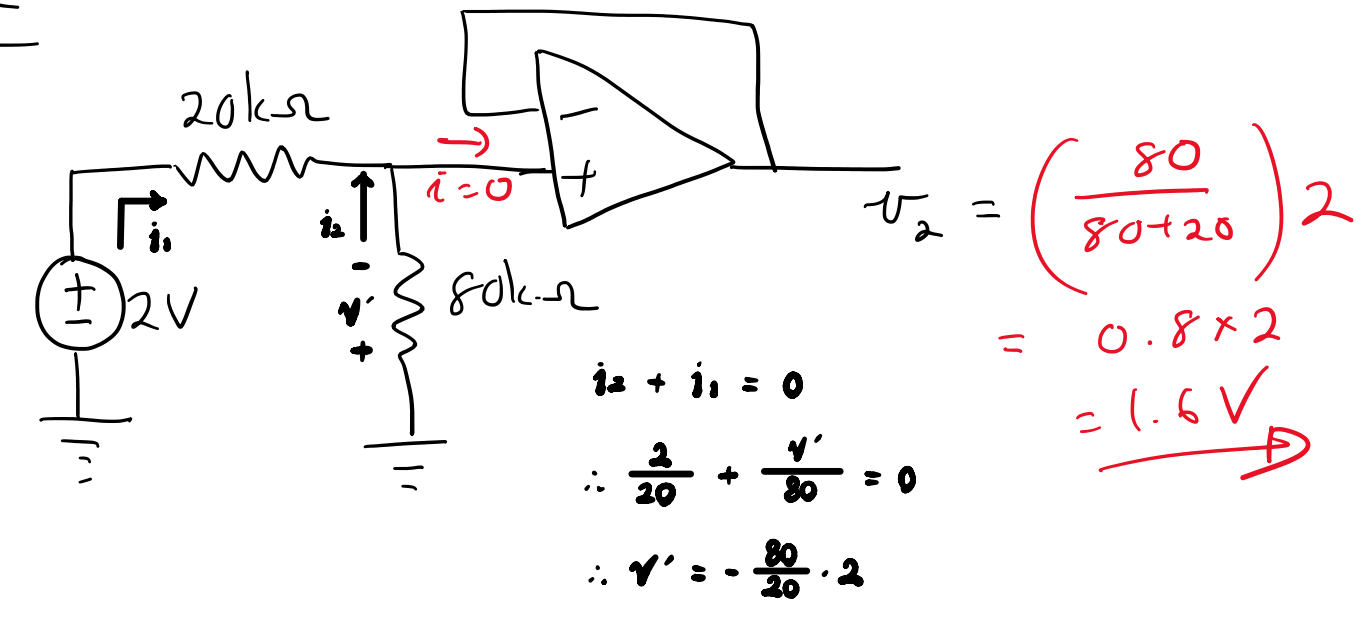


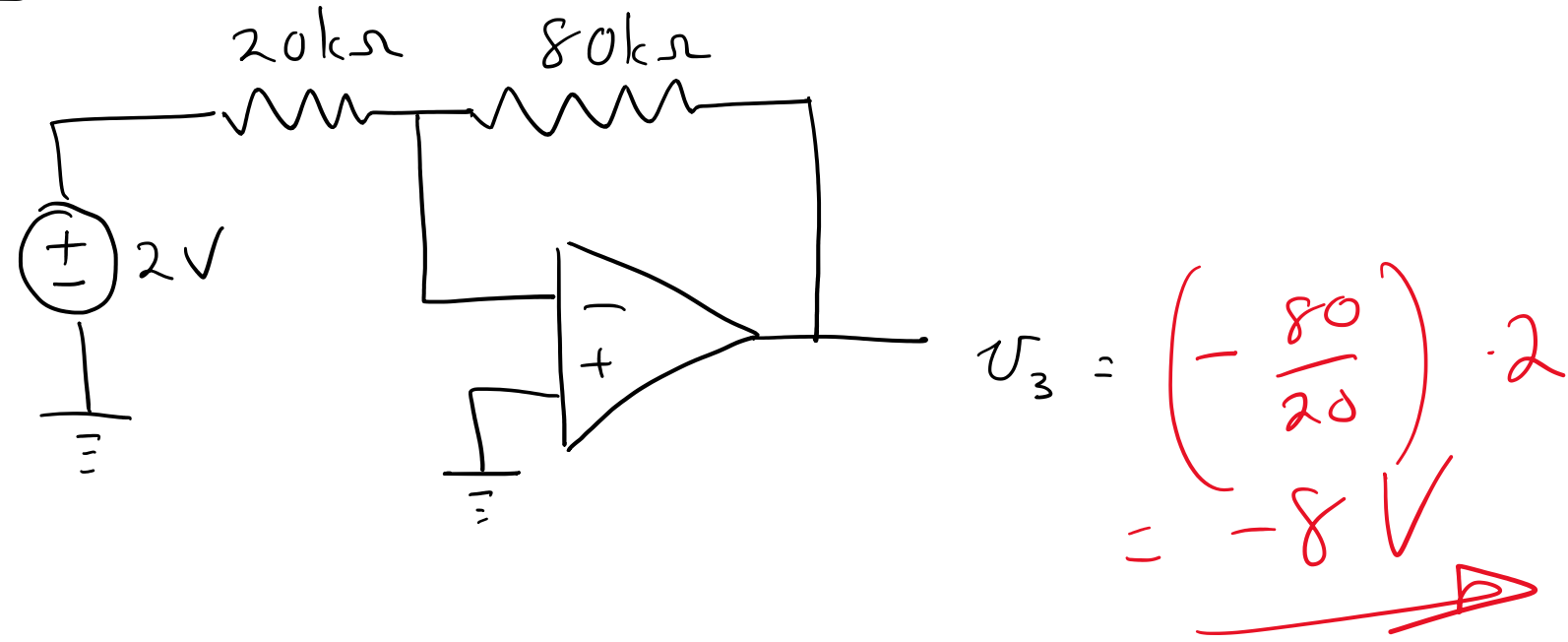
Q1



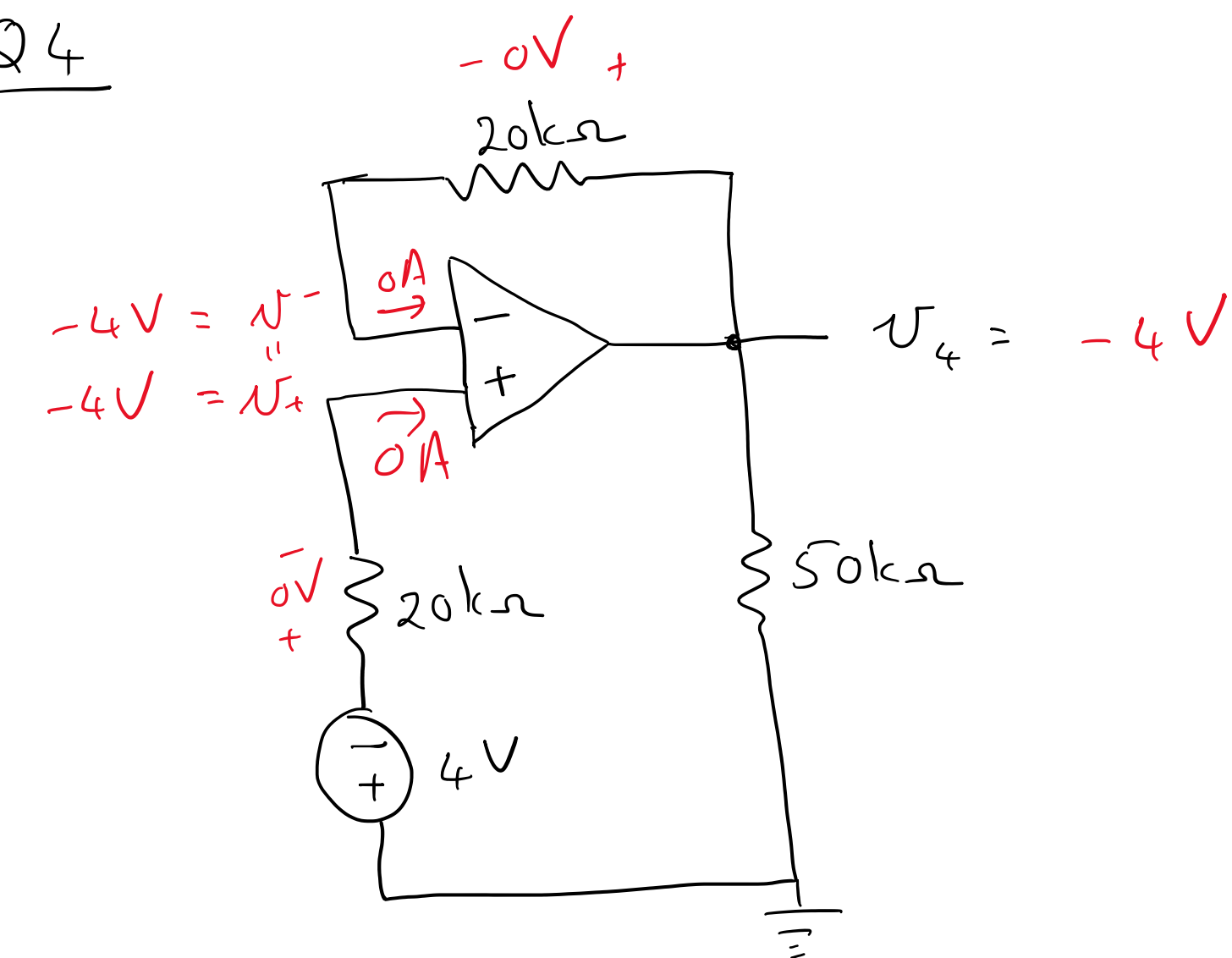
Q2



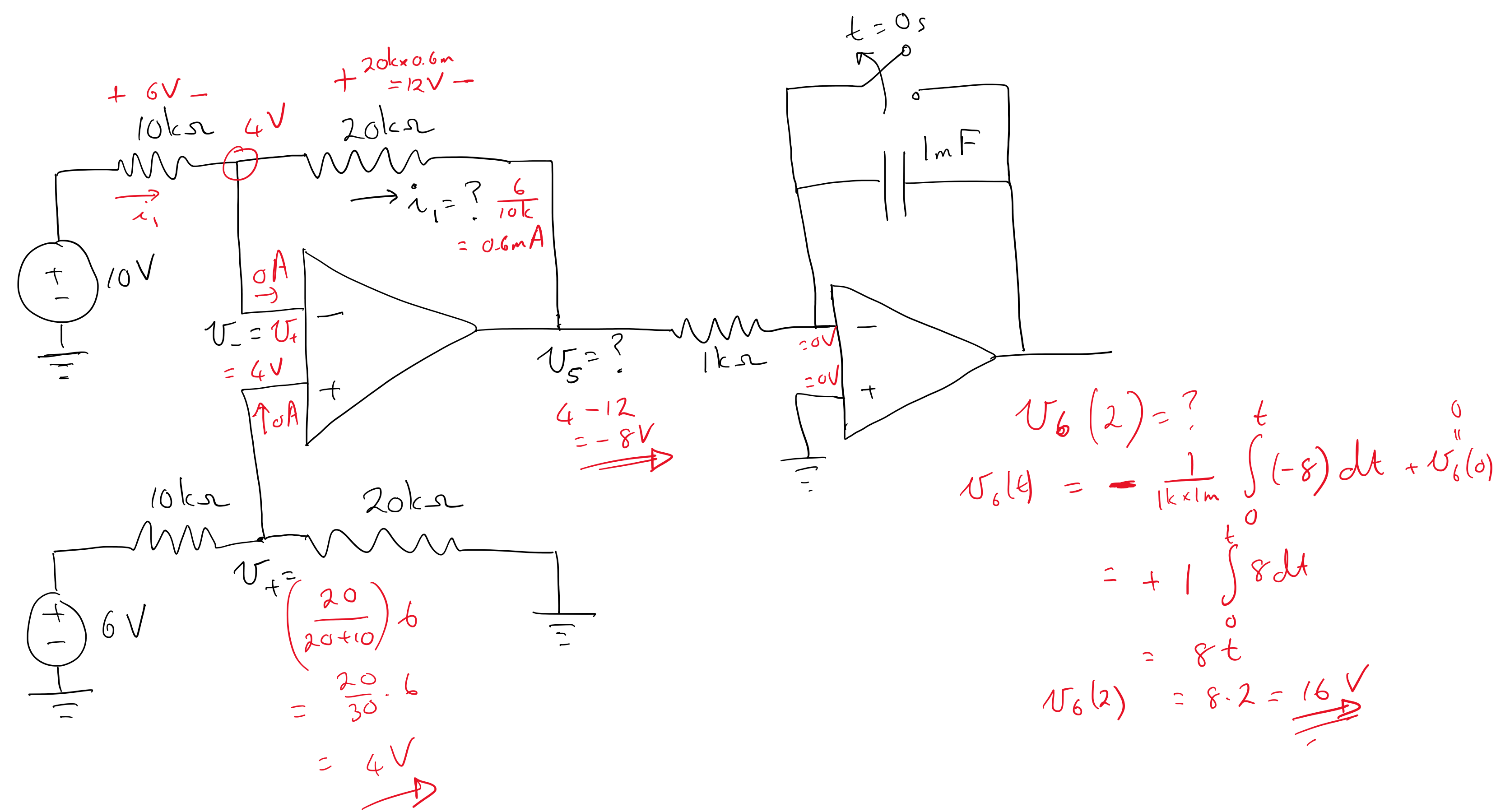
Q3



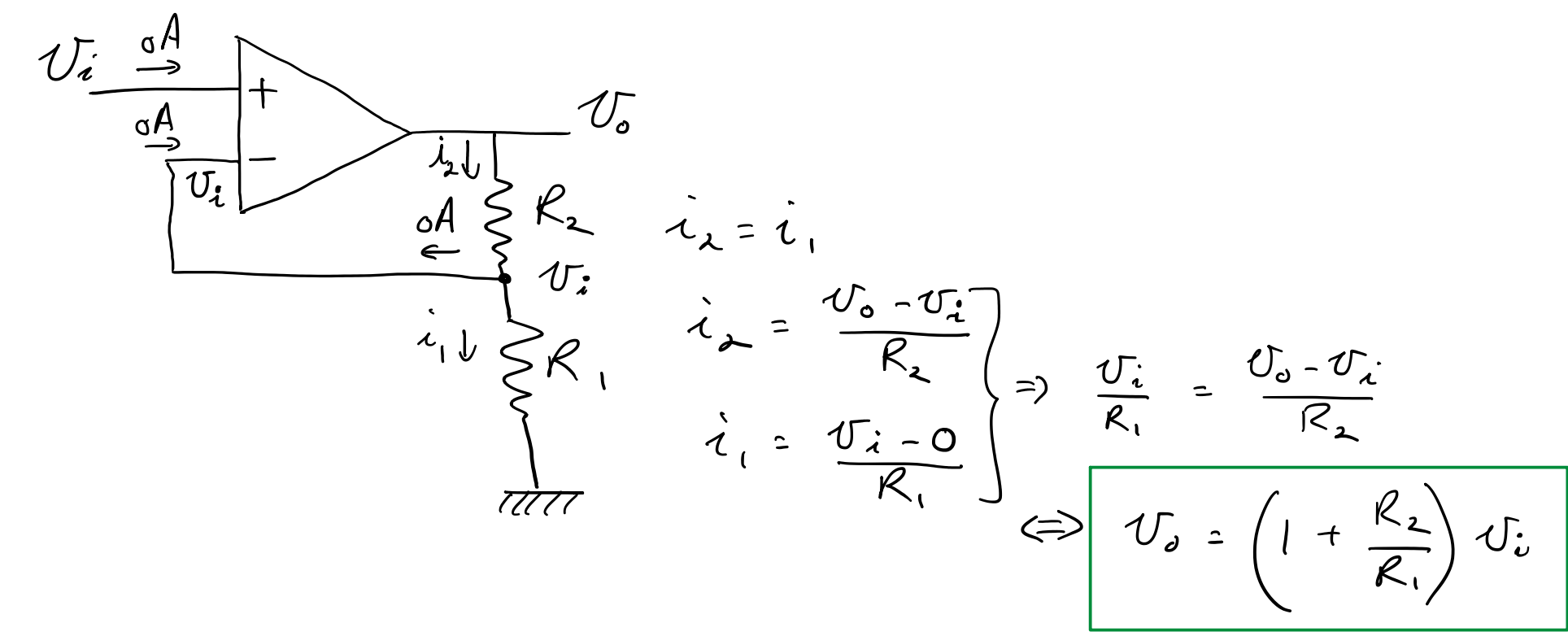
Q4



Q5

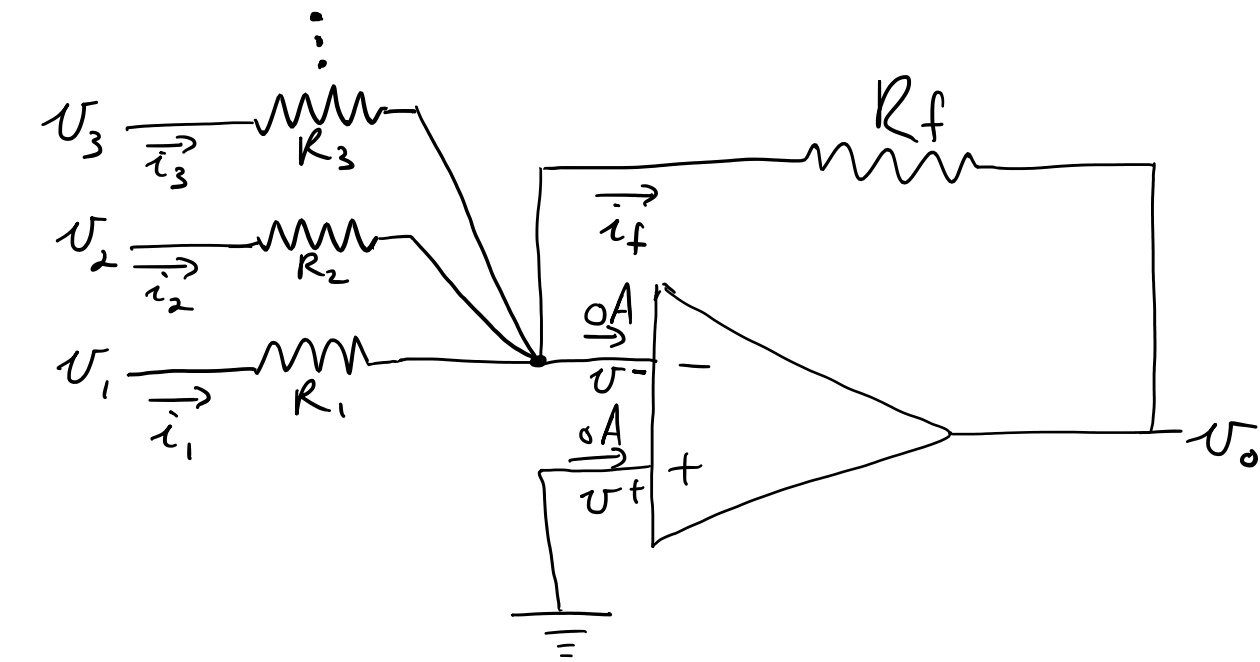


Non-inverting amplifier



$R_2, R_1 > 0 \Rightarrow \text{gain} = 1 + \frac{R_2}{R_1} > 1$
 $\Rightarrow \text{amplifies (cannot attenuate)}$

Inverting summing amplifier



$v^+ = v^-$, but $v^+ = 0V \Rightarrow v^- = 0V$

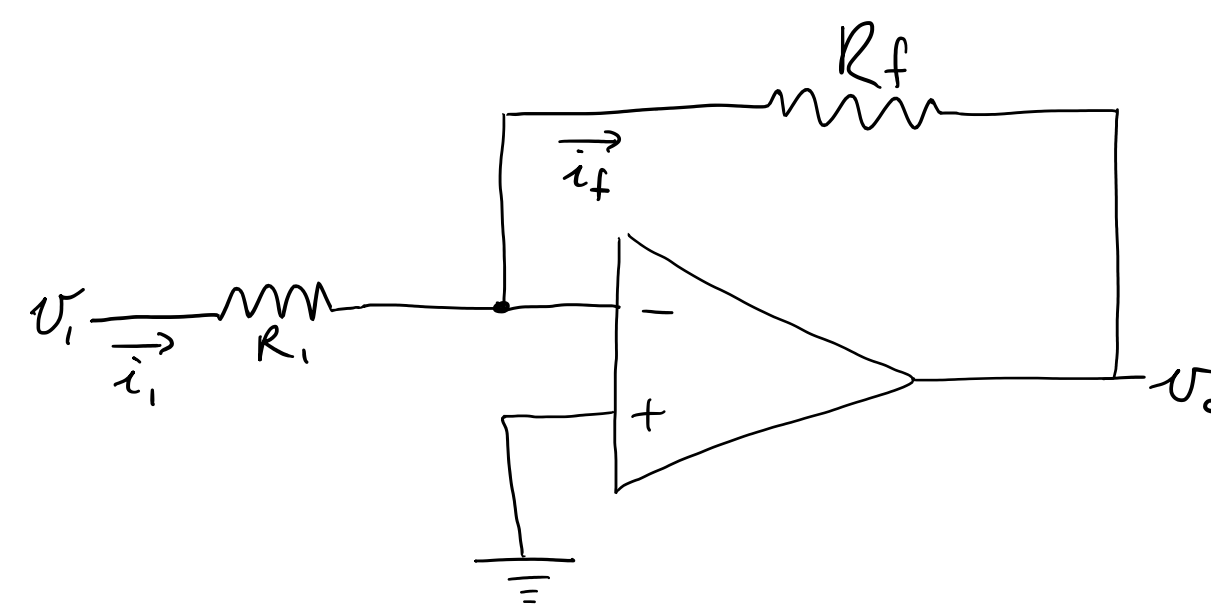
KCL: $i_1 + i_2 + i_3 = i_f + 0$

$$\Leftrightarrow \frac{v_1 - v^-}{R_1} + \frac{v_2 - v^-}{R_2} + \frac{v_3 - v^-}{R_3} = \frac{v^- - v_o}{R_f}$$

$$\Leftrightarrow \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = -\frac{v_o}{R_f}$$

$$\Leftrightarrow v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$

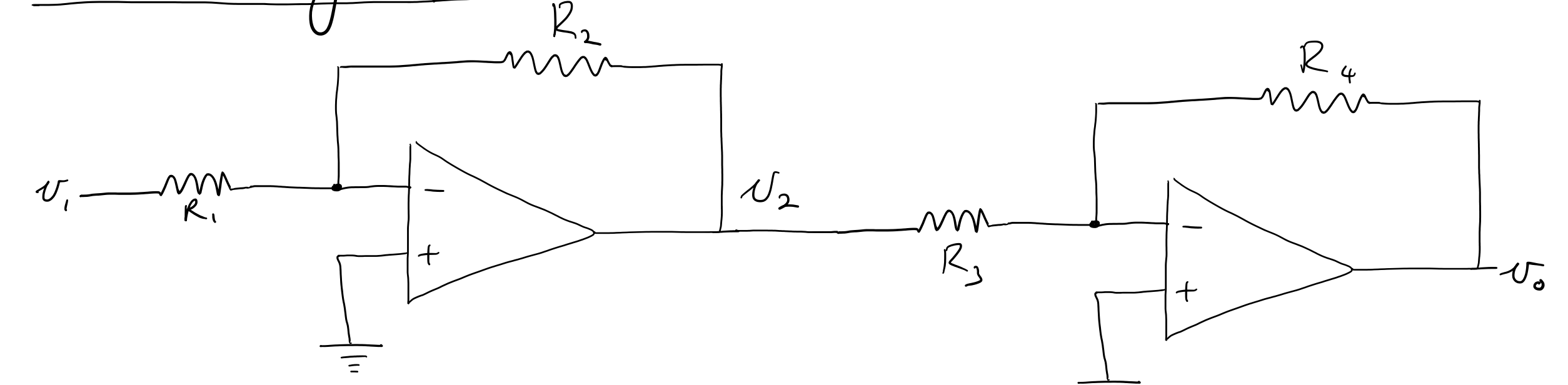
Inverting amplifier



$$v_o = -\frac{R_f}{R_1} v_i$$

$\left|-\frac{R_f}{R_1}\right| > 1 \Rightarrow \text{amplify}$
 $\left|-\frac{R_f}{R_1}\right| < 1 \Rightarrow \text{attenuates}$

Non-inverting amplifier

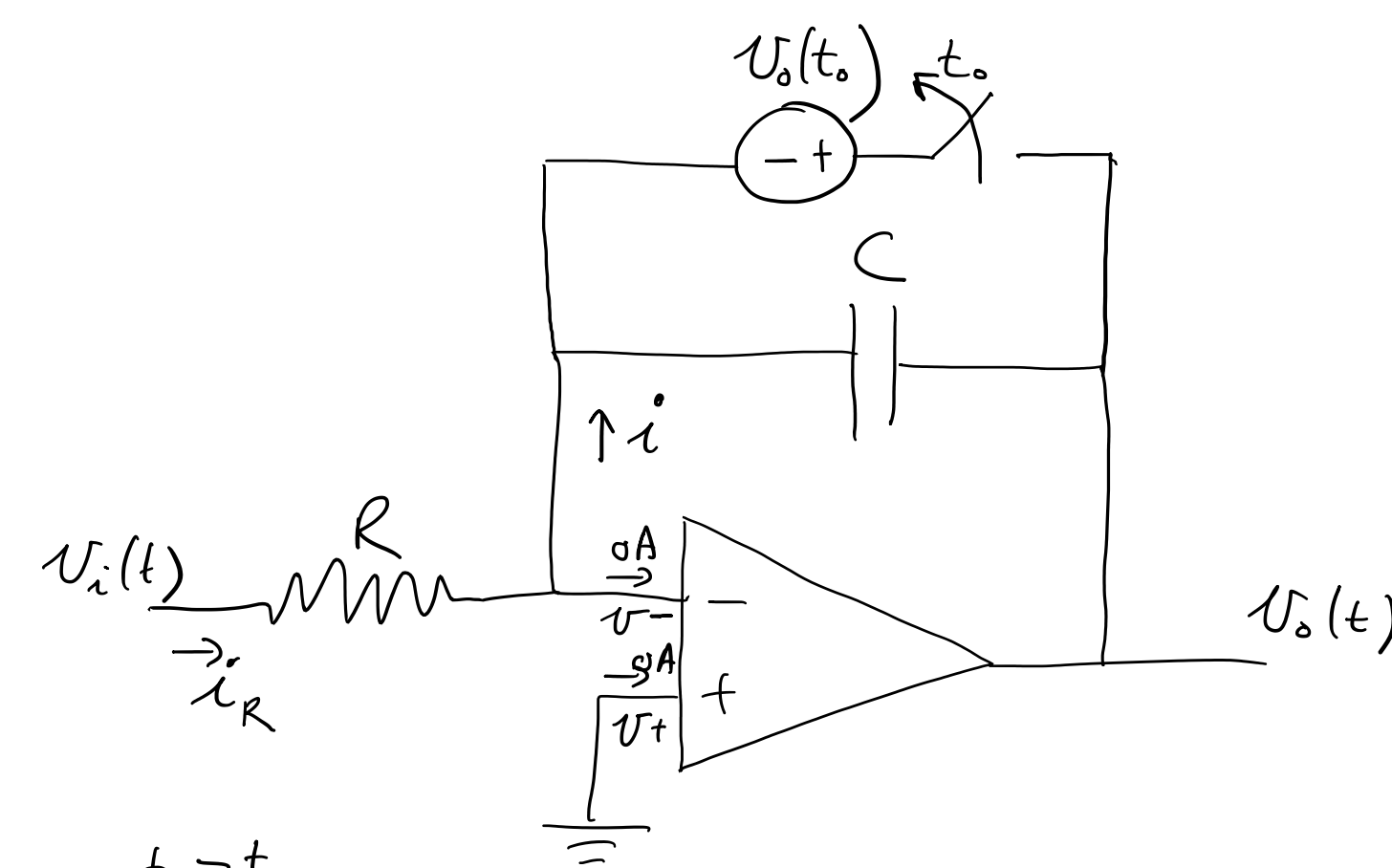


$$v_2 = -\frac{R_2}{R_1} v_1$$

$$v_o = -\frac{R_4}{R_3} v_2 \Rightarrow v_o = \frac{R_4 R_2}{R_1 R_3} v_i$$

can amplify and attenuate

Inverting integrator



$t \geq t_0$
 $v^+ = v^- = 0V$

$$i_2 = i = C \frac{d}{dt} (v^- - v_o(t))$$

$$\Leftrightarrow \frac{v_i}{R} = -C \frac{d v_o(t)}{dt} \Leftrightarrow \frac{d v_o(t)}{dt} = -\frac{1}{RC} v_i(t)$$

$$\Leftrightarrow v_o(t) = v_o(t_0) - \frac{1}{RC} \int_{t_0}^t v_i(\tau) d\tau$$

$$v_i \rightarrow \left[-\frac{1}{RC} \int\right] \rightarrow v_o$$